AI LEGAL DOCUMENT SUMMARIZER

PROJECT WORK
PHASE-I REPORT

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ABSTRACT

The Legal Document Summarizer is an AI-powered system designed to automate the extraction and summarization of complex legal documents using Natural Language Processing (NLP) and deep learning methodologies. This project addresses the critical challenges faced in manual legal document review, including high operational costs, significant time delays, and potential human errors. The system employs transformer-based models fine-tuned on legal corpora to generate both extractive and abstractive summaries with precise clause-level citations.

By integrating Named Entity Recognition (NER) for key information extraction and implementing full traceability through hyperlinked source references, the system transforms legal workflows by reducing manual review time by 50-70% while maintaining accuracy and auditability.

The solution is delivered through a secure web application with a modular architecture comprising a responsive frontend (React.js), RESTful backend API (Flask), and specialized AI engine powered by Hugging Face transformers. This document presents the comprehensive system design, literature survey, technical specifications, and implementation strategy for developing a domain-specific legal summarization platform that ensures quick comprehension, verified summaries, and broad document support across diverse legal document types.

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

NLP - Natural Language Processing

NER - Named Entity Recognition

OCR - Optical Character Recognition

API - Application Programming Interface

REST - Representational State Transfer

DFD - Data Flow Diagram

UI - User Interface

PDF - Portable Document Format

DOCX - Microsoft Word Document Format

TLS - Transport Layer Security

SSL - Secure Sockets Layer

CI/CD - Continuous Integration/Continuous Deployment

BART - Bidirectional and Auto-Regressive Transformers

T5 - Text-to-Text Transfer Transformer

ROUGE - Recall-Oriented Understudy for Gisting Evaluation

F1 - F1 Score (Harmonic Mean of Precision and Recall)

UAT - User Acceptance Testing

RAM - Random Access Memory

ML - Machine Learning

IDE - Integrated Development Environment

CSS - Cascading Style Sheets